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Форма волны или ритм: бенчмарк BPE-токенизации для сигналов ЭЭГ

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Аннотация

Актуальность. Интерфейсы мозг–компьютер (ИМК) и системы автоматического анализа ЭЭГ требуют компактного представления сигнала, инвариантного к межсубъектной вариативности. Методы субсловной токенизации, такие как BPE (Byte Pair Encoding), успешно применяются в обработке естественного языка, однако их потенциал для различных парадигм ЭЭГ систематически не исследован.

Цель исследования: установить условия, при которых BPE-токенизация добавляет ценность для классификации ЭЭГ-парадигм пяти типов (моторное воображение, сон, P300, ментальная арифметика, SSVEP), и разработать практическое правило выбора метода.

Задачи: (1) провести систематический бенчмарк BPE-токенизации на шести наборах данных ЭЭГ с шестью базовыми методами; (2) выполнить аблационные исследования для изоляции вклада метода квантизации, структуры слияний и гиперпараметров; (3) сформулировать правило выбора метода на основе дихотомии амплитудного и частотного кодирования.

Методология. Систематический бенчмарк охватывает шесть публичных наборов данных ЭЭГ (193 испытуемых суммарно): моторное воображение (BCI-IV-2a, 9 исп.; PhysioNet-MI, 109 исп.), определение стадий сна (Sleep-EDF, 20 исп.), вызванные потенциалы P300 (10 исп.), ментальная арифметика (36 исп.) и SSVEP (9 исп.). Квантизованный по амплитуде сигнал кодируется адаптивным BPE и классифицируется тремя вариантами: гистограммы, последовательный CNN и скользящие гистограммы. Сравнение с шестью базовыми методами (CSP+LDA, EEGNet, PSD, VQ и др.) проводится в кросс-субъектной валидации с пятью случайными инициализациями.

Результаты. BPE конкурентоспособен на парадигмах с амплитудным кодированием: Sleep-EDF — 85,1% ($\hat{\kappa}$=0,688) против 85,3% у EEGNet; P300 — 72,6% ($\hat{\kappa}$=0,258); ментальная арифметика — 59,5% ($\hat{\kappa}$=0,189). На парадигмах с частотным кодированием метод уступает: разрыв при моторном воображении достигает $-14{,}5$ п.п. (BCI-IV-2a), а для SSVEP BPE даёт случайный результат (8,7%) против 65,4% у SSVEP-FFT (-57 п.п.). Аблационные исследования подтверждают: структура слияний BPE добавляет $+18{,}0$ п.п. над сырыми бинами и $+5{,}9$ п.п. над случайными слияниями (Sleep-EDF); распределение токенов подчиняется закону Ципфа ($\alpha \approx -1{,}83$, $R^2 \approx 0{,}895$).

Вывод. BPE — эффективный интерпретируемый токенизатор для ЭЭГ-парадигм с амплитудным кодированием, не требующий предобученных моделей. Метод улавливает форму волны, но нечувствителен к модуляции спектральной мощности — сигналу моторного воображения и SSVEP. Предложено практическое правило: BPE для амплитудных парадигм, спектральные методы для частотных.

Ключевые слова: ЭЭГ; BPE-токенизация; интерфейс мозг–компьютер; моторное воображение; определение стадий сна; классификация сигналов; бенчмарк.

Конфликт интересов: авторы декларируют отсутствие явных и потенциальных конфликтов интересов, связанных с публикацией настоящей статьи.

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Waveform or rhythm: BPE tokenisation benchmark for EEG

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Abstract

Relevance. Brain–computer interfaces (BCI) and automated EEG analysis systems require compact signal representations that generalise across subjects. Subword tokenisation methods such as Byte Pair Encoding (BPE) have proven effective in natural language processing, yet their potential for diverse EEG paradigms has not been systematically investigated.

Purpose: To establish the conditions under which BPE tokenisation adds value for EEG classification across five paradigm types (motor imagery, sleep staging, P300, mental arithmetic, SSVEP) and to derive a practical method-selection rule.

Objectives: (1) conduct a systematic benchmark of BPE tokenisation on six EEG datasets against six baselines; (2) perform ablation studies isolating the contribution of quantisation method, merge structure, and hyperparameters; (3) formulate a method-selection rule based on the amplitude/frequency coding dichotomy.

Methods. A systematic benchmark covers six public EEG datasets (193 subjects in total): motor imagery (BCI-IV-2a, 9 subjects; PhysioNet-MI, 109 subjects), sleep staging (Sleep-EDF, 20 subjects), P300 (10 subjects), mental arithmetic (36 subjects), and SSVEP (9 subjects). Amplitude-quantised EEG is encoded with adaptive BPE and classified via histogram, sequence-CNN, and sliding-window histogram variants. Six baselines (CSP+LDA, EEGNet, PSD, VQ, and others) are compared under cross-subject validation with five random seeds.

Results. BPE is competitive on amplitude-coded paradigms: Sleep-EDF 85.1% ($\hat{\kappa}$=0.688) vs. EEGNet 85.3%; P300 72.6% ($\hat{\kappa}$=0.258); mental arithmetic 59.5% ($\hat{\kappa}$=0.189). On frequency-coded paradigms BPE fails: the motor imagery gap reaches -14.5 pp (BCI-IV-2a) and SSVEP BPE yields chance-level accuracy (8.7%) against SSVEP-FFT's 65.4% (-57 pp). Ablation studies confirm that BPE merge structure adds $+18.0$ pp over raw amplitude bins and $+5.9$ pp over random merges (Sleep-EDF); the token frequency distribution follows Zipf's law ($\alpha\approx-1.83$, $R^2\approx0.895$).

Conclusion. BPE is a viable, interpretable tokeniser for amplitude-coded EEG paradigms requiring no large pre-trained model. It captures local waveform shape but is structurally blind to spectral power modulation—the signal exploited by motor imagery and SSVEP paradigms. A practical decision rule is provided: BPE for amplitude-coded paradigms, spectral methods for frequency-coded ones.

Keywords: EEG; BPE tokenisation; brain–computer interface; motor imagery; sleep staging; signal classification; benchmark.

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Introduction

Electroencephalography (EEG) remains the dominant non-invasive modality for brain–computer interfaces (BCIs) owing to its high temporal resolution (on the order of milliseconds), portability and low cost [3, 26, 27]. EEG signals are rich in paradigm-specific information: sleep staging relies on sharp waveform patterns — K-complexes and sleep spindles; P300 interfaces exploit the characteristic positive potential approximately 300 ms after a target stimulus; motor imagery decoding depends on event-related desynchronisation (ERD) of mu and beta rhythms during imagined movement. The diversity of these signals has historically required paradigm-specific features: common spatial patterns (CSP) for motor imagery [4], bandpass filtering for P300 [5], and the fast Fourier transform for SSVEP [6], limiting the generalisability of any single approach. Motor imagery decoding has attracted intensive deep learning research, yet cross-subject generalisation remains a key challenge [27, 28].

A parallel successful direction in natural language processing is *tokenisation*, the conversion of raw text into discrete units whose co-occurrence carries structural meaning. Byte-pair encoding (BPE) [1], the de-facto standard for large language models, iteratively merges the most frequent adjacent symbol pairs in a corpus, forming a vocabulary of sub-word units. The idea is appealing for biosignals. If EEG amplitude is discretised into bins (analogous to characters) and BPE is run, the algorithm may discover recurring waveform patterns (analogous to morphemes) without any task-specific annotation.

A necessary condition for this analogy is that EEG token frequencies obey a power law, as word frequencies do in natural language (Zipf's law [20]). We confirm this empirically on Sleep-EDF data: merged BPE token frequencies follow a power law with exponent $\alpha \approx -1.83$ and $R^2 \approx 0.895$ ($V{=}1024$, $B{=}64$ bins; computed on a 600-epoch subsample from 3 subjects, Fpz–Cz channel; Fig. 1), consistent with the linguistic analogy and suggesting the presence of long-tailed frequency structure in EEG amplitude sequences. The distribution is bimodal: base tokens (ranks 1–64) are approximately equally frequent due to adaptive quantisation, while merged tokens follow the power-law tail.

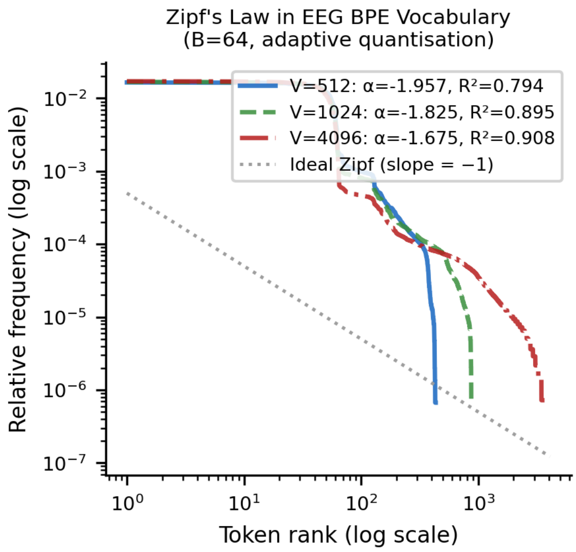

Figure 1: Zipf's law in EEG BPE vocabularies. Token rank vs. frequency on a log–log scale for three vocabulary sizes ($V \in \{512, 1024, 4096\}$, B=64, adaptive quantisation). Fitted exponents (Sleep-EDF, 3 subjects, Fpz–Cz, 600-epoch subsample): $\alpha{\approx}{-1.83}$ (V=1024, R^2=0.895); $\approx{-1.96}$ (V=512, R^2=0.794); $\approx{-1.68}$ (V=4096, R^2=0.908); where α is the log–log slope and R^2 the coefficient of determination. The flat plateau at ranks 1–64 reflects equally-frequent base tokens (adaptive quantisation); the steep tail (merged tokens) exhibits power-law structure comparable to natural language.
Закон Ципфа в BPE-словарях ЭЭГ. Ранг токена против частоты в двойном логарифмическом масштабе для трёх размеров словаря ($V \in \{512, 1024, 4096\}$, B=64, адаптивная квантизация; данные Sleep-EDF). Показатели степенного закона (Sleep-EDF, 3 испытуемых, канал Fpz–Cz, выборка 600 эпох): $\alpha{\approx}{-1{,}83}$ (V=1024, R^2=0,895); $\approx{-1{,}96}$ (V=512); $\approx{-1{,}68}$ (V=4096). Плоское плато на рангах 1–64 соответствует равночастотным базовым токенам (адаптивная квантизация); крутой хвост (слитые токены) подчиняется степенному закону, аналогичному естественному языку.

Related Work

Several works have explored tokenisation for time series and EEG [24]. Chronos [14] applies the language model paradigm to general time series; PatchTST [13] treats fixed temporal patches as tokens; LaBraM [15] pre-trains a transformer on large EEG collections. BIOT [25] unifies biosignal tokenisation from multiple sources. Vector quantisation [12] and contrastive pre-training (BENDR [16]) represent additional directions. EEG-Conformer [21] combines depthwise convolution with transformer attention. Self-supervised learning on raw EEG signals also shows promise [29, 30]. Nevertheless, most works evaluate on one or two paradigms, leaving open the question of *when* tokenisation is applicable.

The answer follows from what BPE actually captures. BPE tokenisation is effective when a paradigm encodes discriminative information primarily in the *waveform* (amplitude domain); it fails when the information lies in *spectral power modulation* (frequency domain). BPE, by merging amplitude-adjacent pairs, is a natural waveform detector, but is structurally insensitive to periodic amplitude modulation. Event-related desynchronisation [11] — the neural mechanism of motor imagery — is precisely such a modulation; sleep staging, by contrast, is an amplitude task.

Research objective. We ask: *under what conditions does BPE tokenisation add value for EEG classification, and why?* This hypothesis, which arose during pilot experiments, is systematically tested across six datasets and five paradigm types with controlled ablations.

Contributions:

1. A systematic benchmark of BPE tokenisation on six public EEG datasets (five BCI paradigm types) with six competing baselines and five random seeds.
2. Empirical establishment of the *amplitude/frequency coding dichotomy* as the primary predictor of BPE effectiveness, supported by controlled ablations.
3. Ablation studies isolating the contribution of quantisation method (A1), spatial integration (A4), BPE merge structure (A7), discriminative merge selection (A8), and the $B\times V$ hyper-parameter grid (A9).
4. Practical guidelines for researchers: when and how to apply BPE tokenisation to EEG signals.

Materials and Methods

Datasets

Six publicly available EEG datasets were selected to maximally cover the diversity of BCI paradigms (Table 1). The selection spans structurally different coding mechanisms: amplitude-modulated waveforms (Sleep-EDF, P300), mixed coding (mental arithmetic, with waveform changes and alpha/theta power shifts), and spectral power modulation (BCI-IV-2a, PhysioNet-MI, SSVEP).

Table 1: Benchmark datasets (N — subjects used in experiments, C — channels, f_s — sampling frequency; *first 20 of 78 available subjects used for computational efficiency; †one of 10 original subjects unavailable in our download).

Dataset	**Paradigm / Парадигма**	N	C	f_s (Hz)	**Epoch / Эпоха**	**Classes / Классы**
BCI-IV-2a [7]	Motor imagery	9	22	250	4 s	4 (L/R hand, feet, tongue)
PhysioNet-MI [8]	Motor imagery	109	64	160	4.5 s	4 (L/R hand, feet, fists)
Sleep-EDF [9]	Sleep staging	20*	2	100	30 s	5 (W, N1, N2, N3, REM)
BNCI P300 [5]	P300 ERP	10	16	2048	1 s	2 (target/non-target)
Mental Arith. [10]	Mental load	36	19	500	60 s	2 (task/rest)
SSVEP [6]	SSVEP	9†	8	256	4 s	12 (frequencies)

A uniform bandpass filter of 0.5–45 Hz (upper cutoff clamped to $f_s/2-1$ Hz when the sampling frequency is low) is applied to all datasets, followed by notch filtering at 50 and 60 Hz. All filters are zero-phase, 4th-order Butterworth. The full BPE tokenisation pipeline is illustrated in Fig. 2.

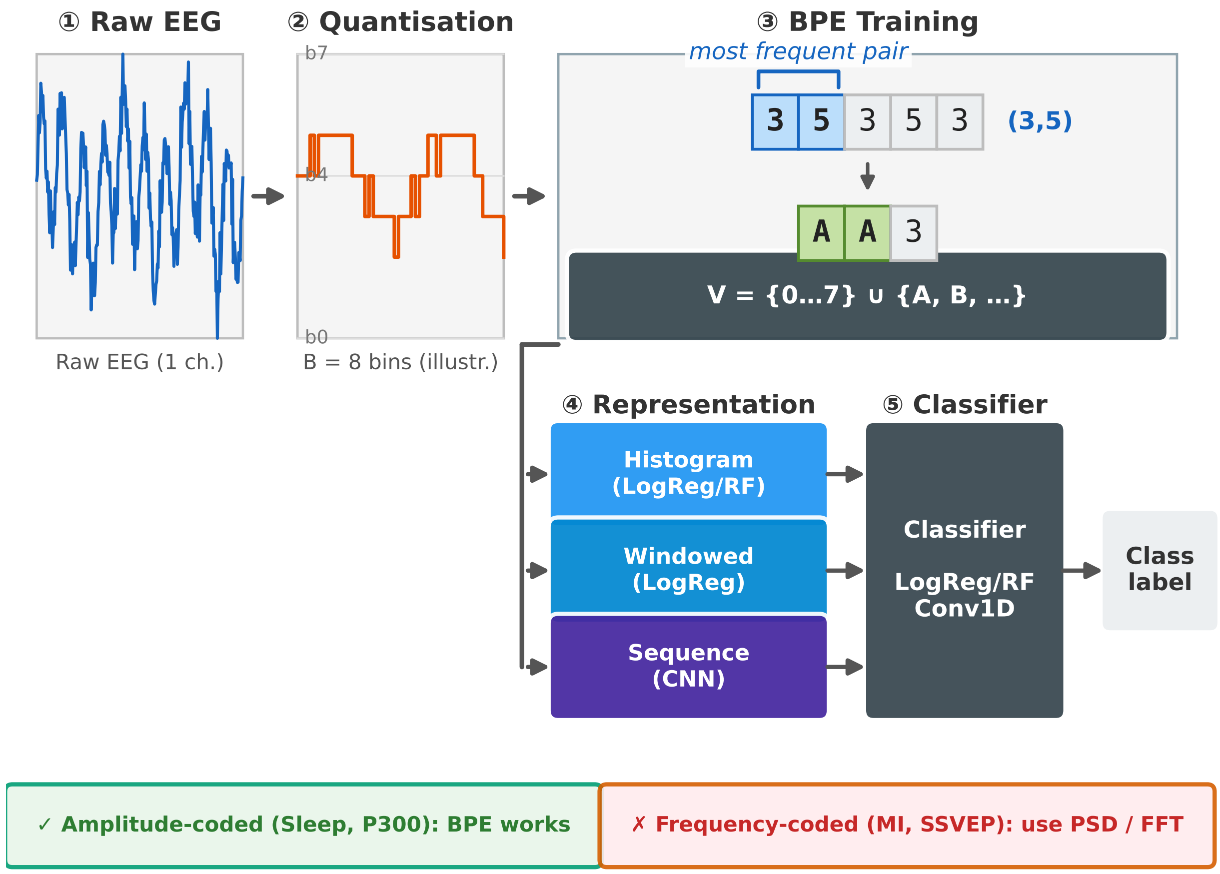

Figure 2: EEG BPE tokenisation pipeline. (1) Raw EEG signal of one channel. (2) Adaptive amplitude quantisation into B bins. (3) BPE training: iterative merging of the most frequent adjacent pair; "AB" is the merged token. (4) Three token representation variants for the classifier. Green/red markers indicate paradigm suitability.
Конвейер BPE-токенизации ЭЭГ. (1) Исходный сигнал ЭЭГ одного канала. (2) Адаптивная квантизация амплитуды по B бинам. (3) Обучение BPE: итеративное слияние наиболее частой смежной пары; «AB» — новый объединённый токен. (4) Три варианта представления токенов для классификатора. Зелёные/красные маркеры указывают применимость к парадигме.

Step 1 — Preprocessing. Each EEG channel is independently filtered, baseline-corrected within the epoch, and downsampled if necessary when $f_s > 500$ Hz. No channel mixing is applied at this stage.

Step 2 — Amplitude quantisation. The continuous amplitude is discretised into B integer bins. *Adaptive quantisation* is used: bin boundaries are set by equi-percentile intervals of the amplitude distribution of the training fold, ensuring uniform occupancy of each bin. This differs from uniform quantisation and μ-law companding; ablation A1 evaluates all three variants. The result is an integer sequence $s = (s_1, \dots, s_T)$, where $s_i \in \{0, \dots, B-1\}$.

Step 3 — BPE vocabulary training. BPE iteratively finds the most frequent adjacent pair (a, b) in the corpus of training sequences and replaces each occurrence with a new merged token [1]. The algorithm is implemented with a doubly-linked list and max-heap with complexity $O(n \log n)$. Training produces a vocabulary of V tokens: B base tokens plus $V - B$ merged tokens. All vocabulary training is performed *exclusively* on the training fold, preventing information leakage from test subjects.

Step 4 — Token representation. Six BPE variants are evaluated in the main benchmark (five representation types, one of which uses two classifiers):

- **Histogram (Hist)**. Relative frequency of each token per channel; feature vector of dimension $V \times C$. PCA is applied to 2048 components, followed by logistic regression or random forest.
- **Windowed (Windowed_LogReg)**. The epoch is split into 250 ms windows, window histograms are concatenated, preserving coarse temporal structure, and a logistic regression classifier is trained.
- **Windowed CNN (Windowed-Seq_CNN)**. The matrix (windows $\times V$) is processed by a 1D convolutional network along the temporal dimension, combining token statistics with sequence dynamics.
- **Sequential CNN (Seq_CNN)**. The raw token sequence is fed to a 1D convolutional network (two Conv1D layers, kernel 3, padding 1, global max pooling, dense layer).
- **Channelwise Transformer**. An independent Transformer encoder with positional encoding per channel, followed by cross-channel pooling.

Baseline Methods

Six baselines represent the main families of EEG features and end-to-end models:

1. **EEGNet** [2]. A compact depthwise-separable CNN for EEG.
2. **CSP+LDA** [4]. Common spatial patterns with linear discriminant analysis; the standard for motor imagery.
3. **PSD_LogReg**. Welch spectral power in five bands ($\delta, \theta, \alpha, \beta, \gamma$) per channel, fed to logistic regression.
4. **VQ_LogReg**. Vector quantisation histogram [12] (k-means codebook on the training fold) with logistic regression.
5. **Patching_LogReg** [13]. Fixed temporal patches with logistic regression.
6. **SSVEP_FFT_LogReg**. FFT power 1–45 Hz with PCA and logistic regression; SSVEP only.

Evaluation Protocol

Cross-validation. For datasets with small subject counts (BCI-IV-2a and SSVEP, 9 subjects each; BNCI P300, 10), *leave-one-subject-out* (LOSO) cross-validation is applied, testing generalisation to completely unseen subjects. For datasets with larger subject pools (PhysioNet-MI, 109 subjects; Sleep-EDF, 20 of 78; Mental Arithmetic, 36), five-fold cross-subject validation is used. In all cases the BPE vocabulary is trained exclusively on the training fold.

Metrics. Accuracy and Cohen's κ [17] are reported. κ is the primary quality indicator for imbalanced datasets (Sleep-EDF has non-uniform sleep stage distribution; P300 has a non-target/target ratio of $\approx$ 6:1). The main benchmark (Table 3) reports *standard* accuracy; ablation tables (A1, A7, A8, A9) report *balanced* accuracy (average recall per class) to highlight per-class performance on imbalanced datasets. The two metrics diverge most on Sleep-EDF: standard accuracy reflects the dominance of stage N2, while balanced accuracy measures performance across all five stages equally.

Reproducibility. All experiments are repeated with five random seeds ($\{42, 123, 456, 789, 2024\}$). Reported values are means over seeds. Implementation uses scikit-learn [18] and PyTorch [19]. Full code, trained vocabularies and raw results are available at `https://github.com/levshaazz/eeg-bpe-benchmark`.

Statistical Analysis

To assess statistical significance of differences between methods, the Wilcoxon signed-rank test [32] is applied to per-fold accuracy vectors for all pairwise comparisons of methods within each dataset. p-values are corrected using the Holm-Bonferroni procedure.

Key comparison results are shown in Table 2. Note that n (the number of folds) is small for all datasets (5 for Sleep-EDF, PhysioNet-MI, Mental Arithmetic; 9 for BCI-IV-2a and SSVEP; 10 for P300), limiting statistical power. No comparison survives correction at the significance threshold α=0.05 (throughout this section α denotes the significance level, distinct from the Zipf exponent in the Introduction). We therefore supplement p-values with Cohen's d_z effect size estimates, which are informative regardless of sample size.

Despite the absence of significant differences after correction, the effect sizes provide a consistent picture. For Sleep-EDF $|d_z| \approx 1.1$–1.2 for BPE_WindowedSeq_CNN vs EEGNet (BPE slightly behind) and EEGNet vs VQ_LogReg (EEGNet marginally ahead on per-fold level; globally VQ=85.9% vs EEGNet=85.3%) indicates a modest spread unlikely to solidify as a systematic advantage with more data. In contrast, for SSVEP $d_z = 2.64$ reflects a fundamental paradigm mismatch rather than sampling variance. For mental arithmetic $d_z = 2.56$ indicates a robust and non-trivial advantage of spectral features over BPE.

Results and Discussion

Main Benchmark

Table 3 and Fig. 3 summarise the full benchmark. Two patterns are clearly apparent.

Table 2: Pairwise statistical comparisons (Wilcoxon signed-rank test). n = number of folds per dataset: 5 for Sleep-EDF, PhysioNet-MI, Mental Arithmetic; 9 for BCI-IV-2a and SSVEP; 10 for P300. Δ — mean accuracy difference (pp); p_{raw} — uncorrected p-value; p_{corr} — Holm-Bonferroni corrected; d_z — Cohen's effect size. No comparison reaches α=0.05 after correction.

Dataset / Датасет	**Method A vs Method B / Методы**	**Δ (pp)**	p_{raw}	p_{corr}	d_z	**Sig.**
Sleep-EDF	BPE_WindowedSeq_CNN vs EEGNet	-3.7	0.125	1.00	-1.20	ns
Sleep-EDF	EEGNet vs VQ_LogReg	$+2.6$	0.125	1.00	$+1.10$	ns
Sleep-EDF	BPE_Hist_RF vs EEGNet	-9.4	0.063	1.00	-1.58	ns
Ment. Arith.	PSD_LogReg vs BPE_Windowed	$+10.7$	0.063	1.00	$+2.56$	ns
Ment. Arith.	VQ_LogReg vs BPE_Windowed	$+8.8$	0.063	1.00	$+1.60$	ns
BNCI P300	EEGNet vs BPE_Hist_RF	$+4.9$	0.002	0.113	$+1.79$	ns
BNCI P300	EEGNet vs BPE_WindowedSeq_CNN	$+5.0$	0.002	0.090	$+2.07$	ns
PhysioNet-MI	EEGNet vs CSP+LDA	$+32.5$	0.063	0.875	$+7.32$	ns
SSVEP	SSVEP_FFT vs BPE_Windowed	$+58.1$	0.004	0.320	$+2.64$	ns

Table 3: Full benchmark: standard accuracy (%) and Cohen's κ. Best BPE result in **bold**; overall best in ***bold italic***. For P300 (highly imbalanced), bold marks the highest-κ BPE variant. Chance: BCI-IV-2a/PhysioNet/SSVEP = 25%/25%/8.3%. Values are mean ± 95% CI over 5 seeds. Ablation tables use balanced accuracy; see Metrics paragraph.

	BCI-IV-2a		**PhysioNet-MI**		**Sleep-EDF**		**P300**		**Ment.**		**SSVEP**	
Method / Метод	Acc.	κ	Acc.	κ	Acc.	κ	Acc.	κ	Acc.	κ	Acc.	κ
BPE variants												
BPE_Hist_LogReg	27.0$_{\pm 0.1}$	.027	32.1	.094	71.9	.506	60.8	.035	**60.5**$_{\pm 1.4}$	**.209**	7.8	$-.006$
BPE_Hist_RF	24.7	$-.004$	26.3	.018	79.5	.497	83.3	.000	54.4	.088	8.6	.003
BPE_Seq_CNN	26.1	.015	26.6	.021	84.2	.665	83.3	.000	50.5	.011	8.4	.001
BPE_Windowed_LogReg	30.2$_{\pm 0.1}$	.069	**42.7**$_{\pm 0.5}$	**.236**	73.0	.537	**72.6**	**.258**	55.9	.117	7.5	$-.009$
BPE_WindowedSeq_CNN	**30.6**$_{\pm 0.5}$	**.074**	37.3	.164	**85.1**$_{\pm 0.1}$	**.688**	83.1	.078	59.5$_{\pm 1.4}$	.189	8.3	$-.001$
BPE_Seq_CW_Transf.	25.0	.000	28.0	.040	76.4	.482	83.4	.052	49.9	$-.003$	8.7	.004
Baselines												
EEGNet	***45.1***$_{\pm 0.8}$	***.268***	***58.2***$_{\pm 1.0}$	***.442***	85.3$_{\pm 2.7}$	.673	***88.0***	***.502***	50.3	.005	8.3	.000
CSP+LDA	39.2	.189	25.5	.006	—	—	—	—	—	—	—	—
VQ_LogReg	28.6	.048	32.3	.096	***85.9***$_{\pm 0.4}$	***.691***	83.3	.000	65.5	.311	8.5	.001
PSD_LogReg	38.0	.174	33.9	.118	80.3	.650	65.6	.130	***68.2***$_{\pm 1.4}$	***.365***	16.7	.091
Patching_LogReg	35.5	.140	25.2	.000	68.5	.000	83.3	.007	50.0	.000	8.3	$-.001$
SSVEP_FFT_LogReg	—	—	—	—	—	—	—	—	—	—	***65.4***	***.623***

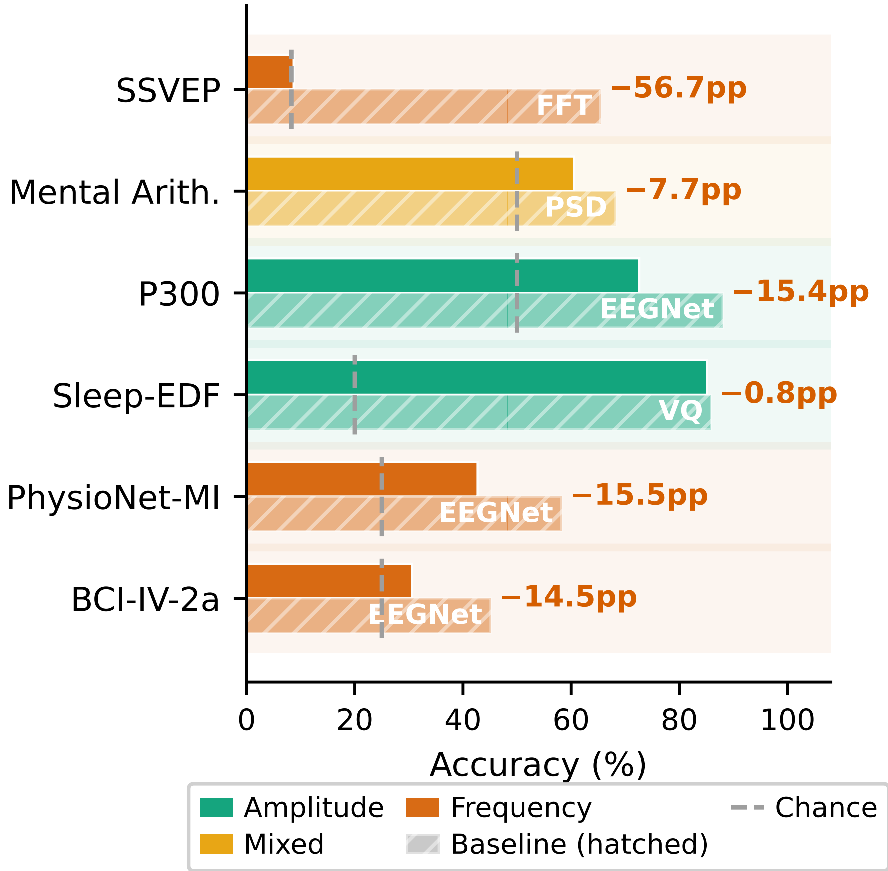

Figure 3: Best BPE variant vs. best baseline per dataset. Dashed line — chance level. Green — amplitude paradigms; orange — mixed; red — frequency.
Лучший BPE против лучшего базового метода. Пунктир — случайный уровень. Зелёный — амплитудные; оранжевый — смешанная; красный — частотные парадигмы.

Amplitude-coded paradigms. On the two amplitude-coded paradigms, Sleep-EDF and P300, BPE variants are highly competitive with EEGNet. BPE_WindowedSeq_CNN achieves 85.1% (κ=0.688) on Sleep-EDF, only 0.2 pp below EEGNet (85.3%). On P300, only BPE_Windowed_LogReg reaches substantial non-chance performance (72.6%, κ=0.258; all other BPE variants yield $\kappa \leq 0.08$). EEGNet (88.0%, κ=0.502) remains the best method for P300, and the gap with BPE in κ is substantial.

The strong Sleep-EDF result is particularly significant because sleep staging is a clinically critical task with heterogeneous signal content. K-complexes and sleep spindles are precisely the local, recurring waveform patterns that BPE detects by design. Published cross-subject SOTA results on comparable Sleep-EDF protocols are 85–87% [22, 23]; our BPE result (85.1%) falls within this range.

Frequency-coded paradigms. On BCI-IV-2a, BPE_WindowedSeq_CNN achieves 30.6% (κ=0.074) versus 45.1% for EEGNet (Δ=−14.5 pp). On SSVEP all BPE variants are at chance level ($\approx$ 8–9%), whereas SSVEP_FFT achieves 65.4%

(κ=0.623). The gap ranges from 56.7 pp (vs. the best BPE variant at 8.7%) to 58.1 pp (vs. BPE_Windowed; Table 2).

Mental arithmetic, a mixed-coding paradigm. Mental arithmetic involves both waveform changes (amplitude component, captured by BPE_Hist_LogReg at 60.5%) and alpha/theta power shifts (frequency component, better captured by PSD_LogReg at 68.2%). Extension K7 tests whether combining both feature sets helps. The BPE+PSD ensemble achieves 68.3% (κ=0.365), identical to PSD alone (68.2%), showing that PSD already captures all the discriminative signal and BPE adds no independent information in this paradigm. Mental arithmetic therefore acts as a *mixed* paradigm; BPE captures the amplitude component but cannot compete with spectral methods because part of the information lies in the frequency domain. This refines the binary amplitude/frequency dichotomy to a three-way classification:

- **Amplitude** (Sleep-EDF, P300). BPE is fully competitive.
- **Mixed** (mental arithmetic). BPE is partially effective.
- **Frequency** (motor imagery, SSVEP). BPE is ineffective.

PhysioNet-MI exception. BPE_Windowed_LogReg achieves 42.7% (κ=0.236) on PhysioNet-MI, ranking second and surpassing CSP+LDA (25.5%, κ=0.006). Ablation A10 (scale control) confirms that dataset size, not paradigm differences, explains this result: when PhysioNet-MI is subsampled to 9 subjects (matching BCI-IV-2a), BPE_Hist_LogReg falls to 28.1% (κ=0.039) — indistinguishable from BCI-IV-2a at the same scale (27.0%, κ=0.027). With all 109 subjects, BPE_Hist_LogReg recovers to 31.9% (κ=0.092), and the windowed variant reaches 42.7% because the larger training set gives the histogram estimator sufficient statistics to capture amplitude envelope modulation that weakly proxies ERD.

Amplitude vs. Frequency Coding Dichotomy

Figures 4 and 5 illustrate the key mechanistic finding for each paradigm type; Fig. 6 provides quantitative confirmation across all six datasets.

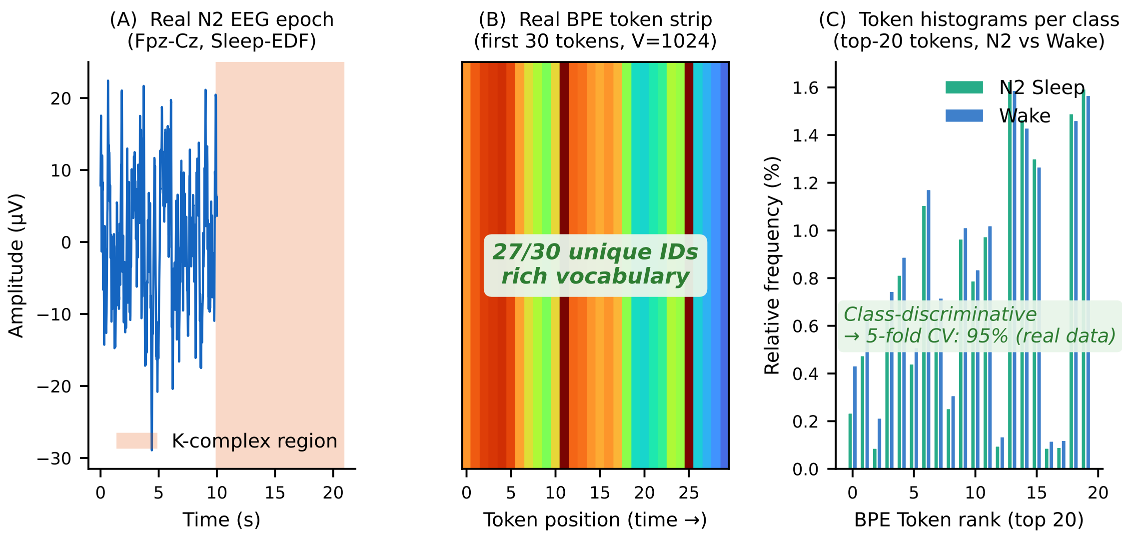

Figure 4: Sleep-EDF — **amplitude** paradigm. **(A)** Real N2 epoch (Fpz–Cz): K-complex highlighted. **(B)** Real BPE token sequence (V=1024): high ID diversity. **(C)** Real class histograms (N2 vs. Wake, 300 epochs each) diverge clearly (5-fold CV: 95%).
Sleep-EDF — **амплитудная** парадигма. **(A)** Реальная эпоха N2 (Fpz–Cz): выделен K-комплекс. **(B)** Реальная последовательность BPE-токенов (V=1024). **(C)** Реальные гистограммы классов (N2 и Бодрствование, по 300 эпох) явно расходятся (5-кратная CV: 95%).

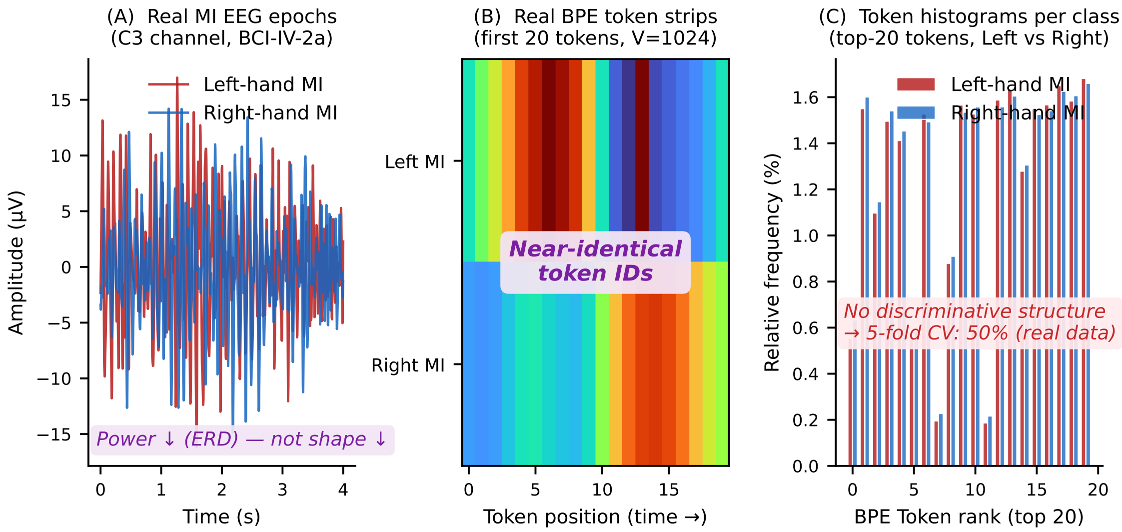

Figure 5: Motor imagery, BCI-IV-2a — **frequency** paradigm. **(A)** Real Left vs. Right MI epochs (C3): ERD reduces power, not waveform shape. **(B)** Real BPE token sequences: nearly identical IDs. **(C)** Real class histograms (Left vs. Right, 200 each) nearly coincide (5-fold CV: 50% $\approx$ chance).
Моторное воображение, BCI-IV-2a — **частотная** парадигма. **(A)** Реальные эпохи Left и Right MI (C3): ERD снижает мощность, но не форму волны. **(B)** Реальные последовательности BPE-токенов: почти идентичные ID. **(C)** Реальные гистограммы классов (Left и Right, по 200) почти совпадают (5-кратная CV: 50% $\approx$ случайный уровень).

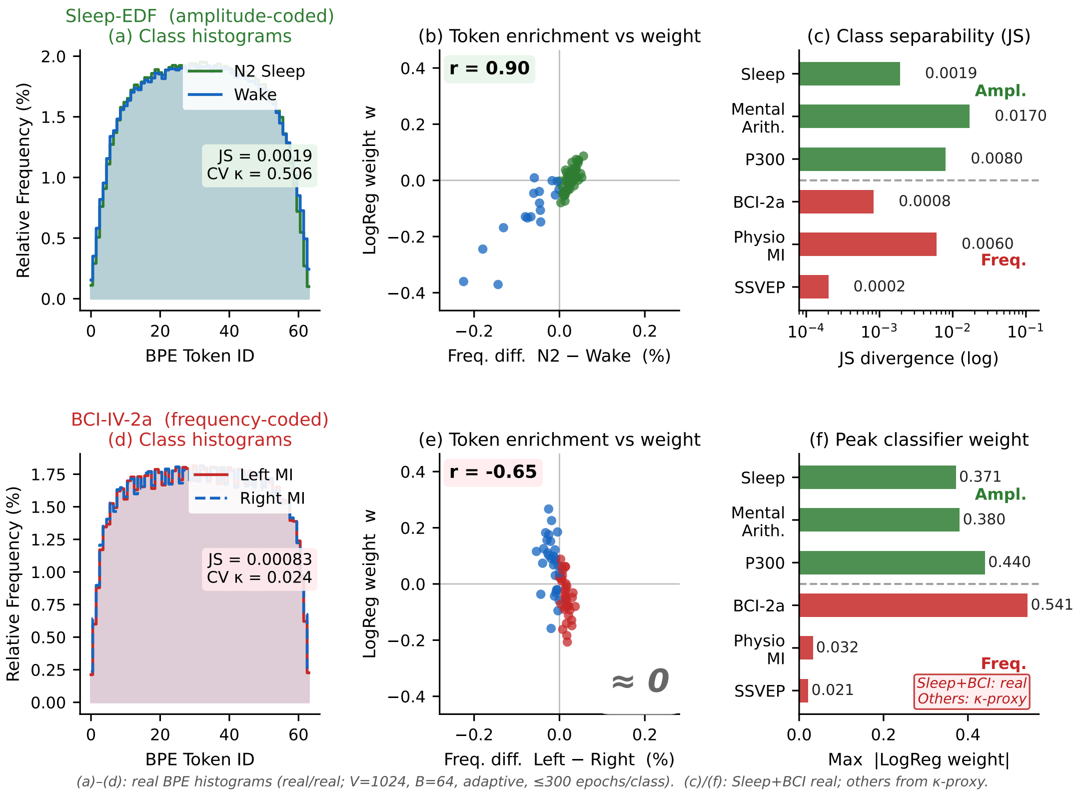

Figure 6: BPE token histogram analysis for the amplitude (Sleep-EDF) and frequency (BCI-IV-2a) paradigms ($V{=}1024$, $B{=}64$, adaptive quantisation, real data). *Panels a, d*: step-line class histograms with JS divergence and cross-validated κ (Sleep $\kappa{=}0.506$; BCI $\kappa{=}0.024$). *Panels b, e*: scatter of per-token frequency difference vs. LogReg weight — Sleep shows strong correlation ($r{=}0.90$), BCI collapses to the origin. *Panels c, f*: JS divergence and peak |LogReg weight| across all six datasets (Sleep and BCI: real; others: κ-proxy).

Анализ гистограмм BPE-токенов для амплитудной (Sleep-EDF) и частотной (BCI-IV-2a) парадигм ($V{=}1024$, $B{=}64$, адаптивная квантизация, реальные данные). *Панели a, d*: ступенчатые гистограммы классов с дивергенцией Дженсена–Шеннона и кросс-валидационным κ (Sleep $\kappa{=}0{,}506$; BCI $\kappa{=}0{,}024$). *Панели b, e*: скаттерплот частотного различия токенов и весов LogReg — Sleep демонстрирует сильную корреляцию ($r{=}0{,}90$), BCI схлопывается к началу координат. *Панели c, f*: дивергенция Дженсена–Шеннона и пиковый |вес LogReg| по всем шести наборам данных.

Figure 6 panels a, d show real BPE token histograms for Sleep-EDF (CV $\kappa{=}0.506$) and BCI-IV-2a ($\kappa{=}0.024$); panels b, e reveal that per-token enrichment strongly predicts classifier weight for amplitude-coded data ($r{=}0.90$) but not for frequency-coded data; panels c, f summarise Jensen-Shannon divergence and peak classifier weight across all six datasets.

BPE builds its vocabulary by merging *adjacent amplitude patterns* and therefore captures *local waveform*, precisely the structure exploited by sleep staging and P300 classification. It is structurally insensitive to *periodic amplitude modulation at a specific frequency*. A 20 Hz sinusoid quantised into B bins produces a highly regular token sequence whose histogram is practically identical to that of a 16 Hz sinusoid with suppressed amplitude. ERD [11], the reduction of mu/beta *oscillatory* power during motor imagery, is precisely such periodic modulation; harmonic resonance in SSVEP is even more pronounced.

Ablation Studies

A1: Quantisation method

Adaptive quantisation outperforms uniform and μ-law on Sleep-EDF (59.7% vs. 59.0% and 55.7%), while differences on BCI-IV-2a are within ± 1 pp (Table 4). All subsequent experiments use adaptive quantisation.

Table 4: Ablation A1: quantisation method (BPE_Hist_LogReg, V=4096, B=64; balanced accuracy). Абляция A1: метод квантизации (BPE_Hist_LogReg, V=4096, B=64; balanced accuracy).

Quantisation / Квантизация	Sleep-EDF (%)	BCI-IV-2a (%)
Adaptive (percentiles)	**59.7**	28.0
Uniform	59.0	27.4
μ-law	55.7	**28.6**

A4: Spatial integration

Standard BPE processes each EEG channel independently. Additionally, *Spatial BPE* was evaluated: k-means clustering of multi-channel amplitude snapshots followed by BPE on the resulting spatial code sequences. On Sleep-EDF, Spatial_BPE_RF achieves κ=0.693, matching the best channel-independent variant (BPE_WindowedSeq_CNN, κ=0.688), indicating that spatial pooling of quantised snapshots does not capture additional discriminative structure beyond individual channels. The per-fold codebook design (fitting k-means exclusively on training data) prevents data leakage; a pilot with a global codebook inflated Sleep accuracy by +29 pp — a spurious gain attributable entirely to leakage. On motor imagery the method is similarly ineffective — consistent with the frequency-coding bottleneck.

A7: Does the BPE merge structure matter?

Three conditions are compared on Sleep-EDF and BCI-IV-2a (Fig. 7, Table 5). **BPE** uses the standard merge order, **Random** uses a random merge order at the same vocabulary size, and **Raw bins** uses no merges (base bins only, $V = B$). Each condition uses a *freshly trained* vocabulary, introducing higher seed-to-seed variance than ablations that reuse the main-experiment vocabulary (see footnote in A8).

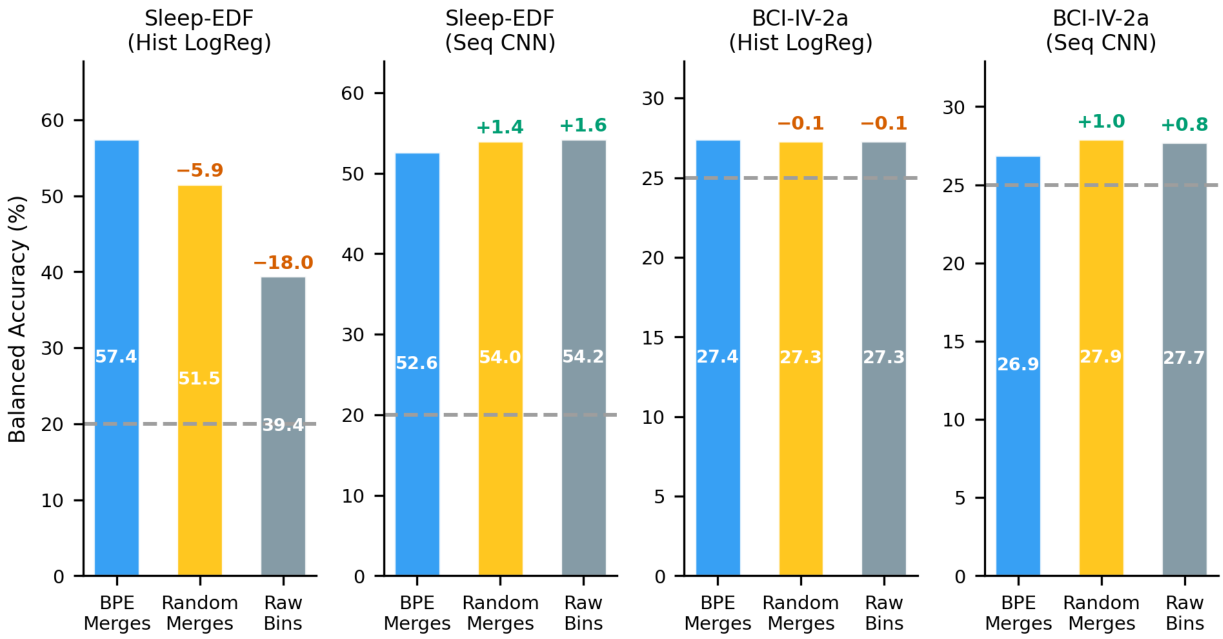

Figure 7: Ablation A7: BPE merge structure vs. random merges vs. raw bins for histogram and CNN classifiers. Marked deltas are relative to BPE. Dashed line — chance level.
Абляция A7: структура слияний BPE против случайных слияний и сырых бинов для гистограммного и CNN-классификаторов. Отмеченные дельты — относительно BPE. Пунктирная линия — уровень случайного угадывания.

Table 5: Ablation A7: balanced accuracy (%).

Dataset / Датасет	Classif. / Классиф.	BPE	Rand. / Случ.	Raw / Сырые
Sleep-EDF	Hist_LR	**57.4**	51.5	39.4
Sleep-EDF	Seq_CNN	52.6	54.0	**54.2**
BCI-IV-2a	Hist_LR	27.4	27.3	27.3
BCI-IV-2a	Seq_CNN	26.9	**27.9**	27.7

On Sleep-EDF **histograms**: BPE adds $+18.0$ pp over raw bins and $+5.9$ pp over random merges, confirming that the *order* of merges carries genuine information for the histogram representation. However, on Sleep-EDF **Seq_CNN** all three conditions are statistically indistinguishable (52.6% vs. 54.0% vs. 54.2%): the convolutional classifier recovers temporal structure directly from the token sequence, rendering the specific merge order irrelevant. On BCI-IV-2a all conditions are indistinguishable and close to chance level — consistent with our prediction.

A8: Discriminative BPE

Standard BPE selects merges by frequency alone, ignoring class labels. Discriminative merge selection was evaluated using the score $\mathrm{score}(a,b) = (1-\lambda)\cdot \mathrm{frequency} + \lambda \cdot \mathrm{Fisher\ ratio}$, where the Fisher ratio measures class separability and $\lambda \in [0,1]$ controls the blend (not to be confused with the significance level α=0.05 used in Section *Statistical Analysis*). Values $\lambda \in \{0, 0.5, 1.0\}$ were tested on BCI-IV-2a and Sleep-EDF.

Results are consistently negative: on BCI-IV-2a all λ values yield $\approx$ 27–28%; on Sleep-EDF increasing

[1]The A8 baseline (Standard BPE, λ=0) reports 62.2% on Sleep-EDF, while the nominally identical BPE condition in A7 reports 57.4%. Both ablations use V=4096 and balanced accuracy. The ≈5 pp discrepancy likely reflects

λ from 0 to 1.0 *reduces* accuracy from 62.2% to 58.5% (-3.7 pp).[1]

A9: $B \times V$ hyperparameter grid

Figure 8 shows the joint effect of the number of bins B and vocabulary size V on Sleep-EDF and BCI-IV-2a.

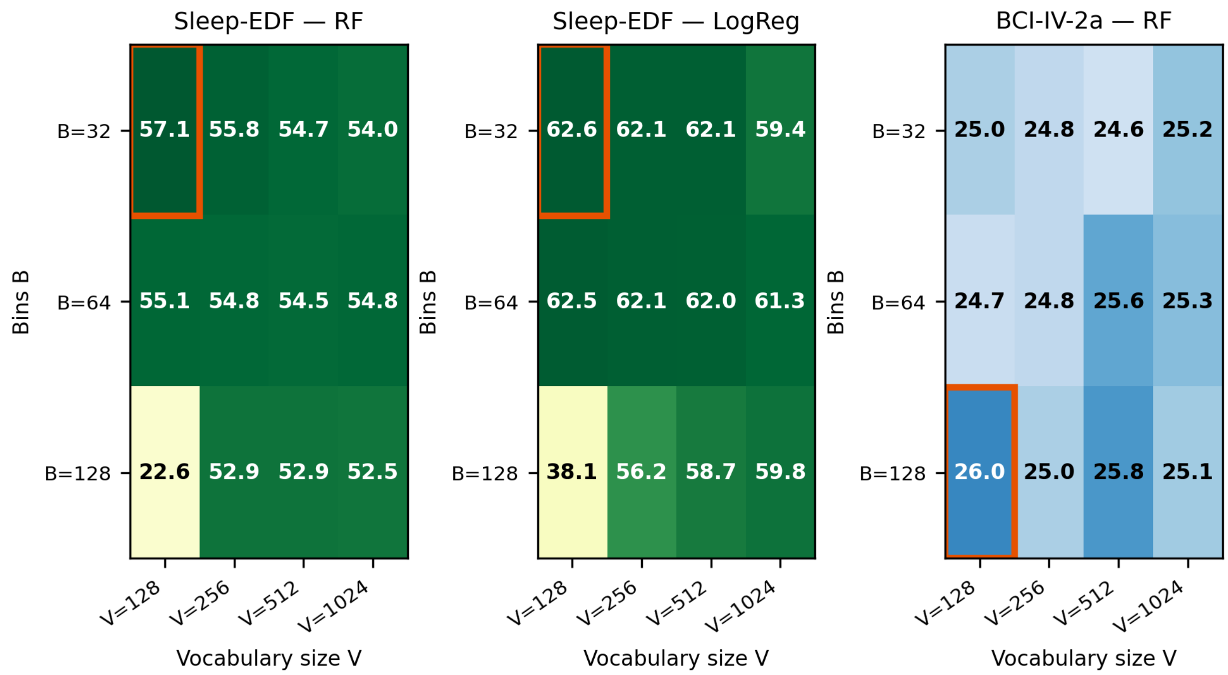

Figure 8: Ablation A9: accuracy (%) for the $B \times V$ grid. Orange border — optimum. Sleep-EDF RF: smaller B and V are consistently better. BCI-IV-2a: all configurations are near chance.
Абляция A9: точность (%) для сетки $B \times V$. Оранжевая рамка — оптимум. Sleep-EDF RF: меньшие B и V стабильно лучше. BCI-IV-2a: все конфигурации вблизи случайного уровня.

On Sleep-EDF with random forest, the optimum within the tested grid is $B{=}32$, $V{=}128$ (57.1%); balanced accuracy generally decreases as B or V grows within $V{\leq}1024$. (The cell $B{=}128$, $V{=}128$ collapses to 22.6% because $V{=}B$ implies zero merges, yielding a degenerate raw-bin histogram.) On BCI-IV-2a *no* combination of B and V exceeds 28% — the paradigm limitation acts uniformly.

Extensions: CSP+BPE, Spatial BPE, and BPE+PSD Ensemble

Table 6 summarises three extensions evaluated beyond the main benchmark.

CSP+BPE. Common spatial patterns project the multi-channel signal onto components that maximise the variance difference between classes. Applying CSP before BPE partially converts ERD (a spectral phenomenon) into amplitude variance of the spatial compo-

higher seed-to-seed variance in A7 (where vocabularies are retrained for each merge-order condition), as opposed to A8 (which reuses the main-experiment vocabulary). The main conclusions of both ablations are unaffected.

Table 6: Extension methods: accuracy (%). † chance = 50% for mental arithmetic. K7 experiments use V=1024; main benchmark uses V=4096.

Method / Метод	BCI-IV-2a	PhysioNet-MI	Sleep-EDF	Mental Arith.† / Мент. ариф.
CSP+BPE_LogReg	31.6	24.8	—	—
Spatial_BPE_RF	—	33.7	85.9	—
BPE_Hist_LogReg (K7)	—	—	—	62.9
PSD_LogReg (K7)	—	—	—	68.2
BPE+PSD_Ensemble (K7)	—	—	—	68.3
Best BPE (main)	30.6	42.7	85.1	60.5

nents, which BPE can then exploit. CSP+BPE_LogReg achieves 31.6% on BCI-IV-2a — a modest $+1.0$ pp over BPE_WindowedSeq_CNN (30.6%), but still 8 pp below CSP+LDA (39.2%). On PhysioNet-MI with 109 subjects, CSP+BPE is near chance level, indicating that the CSP bridge is insufficient at large scale and that the dominant ERD/ERS information remains spectral, not amplitude-coded.

Spatial BPE. K-means clustering of multi-channel amplitude snapshots (k=128) with BPE on the resulting spatial code sequence yields Spatial_BPE_RF with κ=0.693 on Sleep-EDF, comparable to channel-independent BPE_WindowedSeq_CNN (κ=0.688). Spatial tokenisation discards within-channel temporal structure, which is the primary discriminative signal for amplitude-coded paradigms. On motor imagery the method is also ineffective, consistent with the amplitude–frequency dichotomy.

BPE+PSD Ensemble (K7). For the mixed-coding mental arithmetic paradigm, we test whether BPE and PSD features capture complementary information. BPE_Hist_LogReg achieves 62.9% (κ=0.258, V=1024); PSD_LogReg achieves 68.2% (κ=0.365). Their concatenated ensemble reaches 68.3% (κ=0.365) — a +0.1 pp gain indistinguishable from noise. The PSD features subsume the BPE signal: once band-power is available, the token co-occurrence statistics add no independent information. Practitioners working with mixed-coding paradigms should therefore apply PSD directly rather than constructing a hybrid feature set.

Alternative Tokenisation Approaches

Beyond raw amplitude quantisation, five alternative preprocessing strategies prior to BPE tokenisation were evaluated, each targeting different signal components (Table 7, Fig. 9).

Table 7: Alternative tokenisation approaches: accuracy (%) for Sleep-EDF and BCI-IV-2a. Best approach per dataset in **bold**.

Approach / Подход	Sleep-EDF	BCI-IV-2a
Raw amplitude (base)	80.9	24.4
Delta quantisation	79.4	25.3
Downsample 64 Hz	75.9	27.1
Envelope (8–30 Hz)	78.9	25.7
Spectral 500 ms	**81.8**	26.0
Multiband ($\theta/\alpha/\beta$)	73.2	26.4

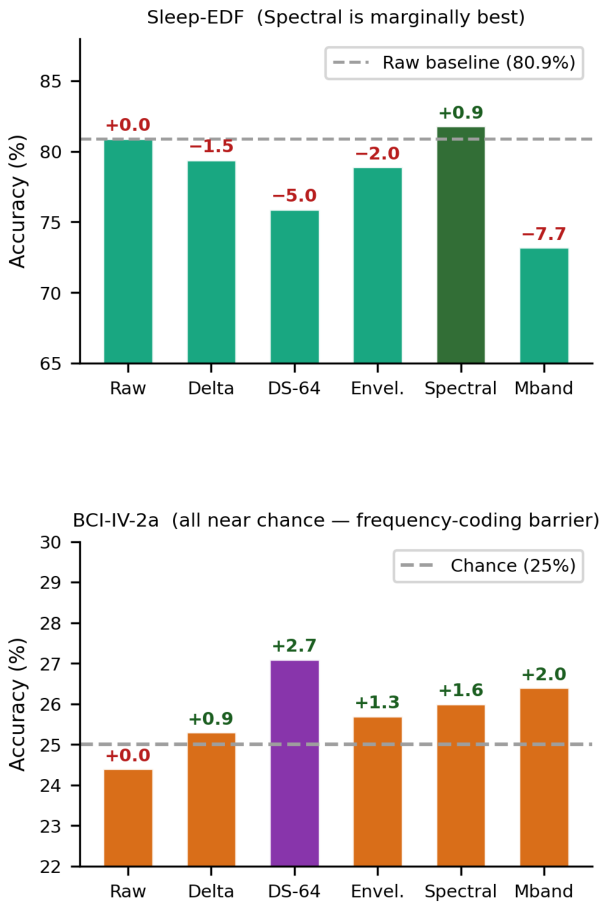

Figure 9: Alternative tokenisation approaches (Experiment 6). Deltas relative to raw amplitude baseline. DS-64 = Downsample 64 Hz; Envel. = Envelope 8–30 Hz; Mband = Multiband $\theta/\alpha/\beta$. *Upper*: Sleep-EDF — spectral tokenisation is marginally better ($+0.9$ pp over raw). *Lower*: BCI-IV-2a — all approaches near chance level. Альтернативные подходы к токенизации (Эксперимент 6). Дельты относительно базового варианта с сырой амплитудой. DS-64 = понижение частоты до 64 Гц; Mband = мультиполосная огибающая. *Сверху*: Sleep-EDF — спектральная токенизация незначительно лучше ($+0{,}9$ п.п. над базовым вариантом). *Снизу*: BCI-IV-2a — все подходы вблизи случайного уровня.

For Sleep-EDF, spectral tokenisation (STFT magnitude bins) is marginally best ($+0.9$ pp), while multiband envelope concatenation is the worst (-7.7 pp). For BCI-IV-2a all approaches remain at chance level (24–27%), confirming that no tokenisation-level preprocessing is able to overcome the frequency-coding barrier. The raw amplitude baseline (80.9% Sleep-EDF) uses a histogram LogReg classifier evaluated in a separate cross-validation; the higher values in Table 3 reflect the RF and sequence-CNN classifiers that use the merged BPE vocabulary.

Token Interpretability

To assess the correspondence of BPE tokens to known neurophysiological events, a binary logistic regression (N2 vs. Wake) was trained on BPE histogram features from Sleep-EDF (N=7117 epochs; B=64, V=1024). Tokens were ranked by a composite score $s = w \cdot \log(1 + \ell)$, where w is the LogReg coefficient and ℓ is the token length in bins.

For each selected token, *event-triggered averaging* (ETA) was computed: all occurrences of the token in epochs of the corresponding class were localised with position tracking; a ±600 ms raw EEG window around the token centre was extracted and averaged across all occurrences ($n \geq 100$ per token). Figure 10 presents the ETA signals alongside canonical biomarkers from the literature.

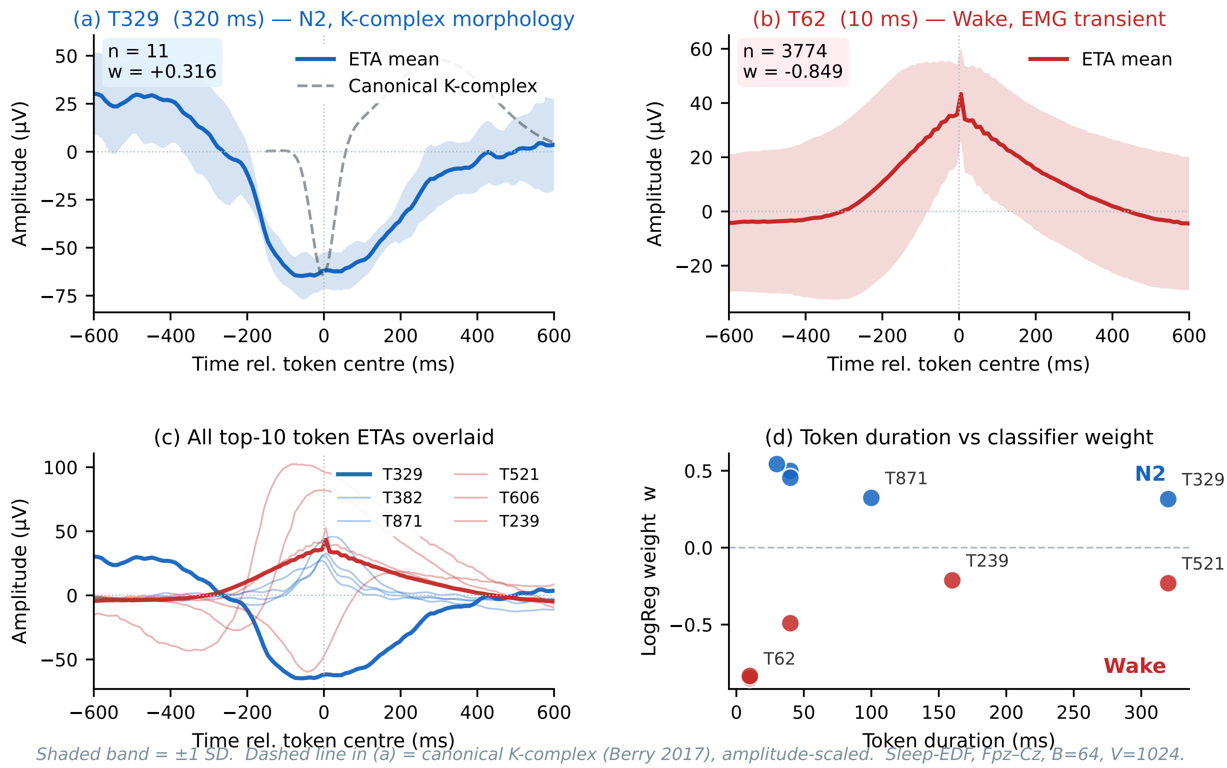

Figure 10: BPE token interpretability via event-triggered averaging (ETA), Sleep-EDF (N2 vs. Wake, Fpz–Cz, B=64, V=1024; the main benchmark uses V=4096). *Panel (a)*: token T329 (ℓ=32 bins, 320 ms) ETA with canonical K-complex overlay (dashed); the negative deflection matches K-complex morphology. *Panel (b)*: token T62 (ℓ=1 bin, 10 ms, w=−0.849) — a single-bin base token whose ETA shows a sharp positive spike consistent with EMG/movement transients. *Panel (c)*: all 10 selected token ETAs overlaid (blue = N2, red = Wake). *Panel (d)*: token duration vs. classifier weight, showing that the most discriminative tokens span a wide range of lengths (10–320 ms).

Интерпретируемость BPE-токенов посредством усреднения по событиям (ETA), Sleep-EDF (N2 против Бодрствования, Fpz–Cz, B=64, V=1024; в основном бенчмарке V=4096). *Панель (a)*: токен T329 (ℓ=32 бина, 320 мс) ETA с наложением канонического K-комплекса (пунктир). *Панель (b)*: токен T62 (ℓ=1 бин, 10 мс, w=−0,849) — однобиновый базовый токен, ETA которого показывает острый пик, согласующийся с ЭМГ-артефактами. *Панель (c)*: все 10 выбранных токенов наложены (синий = N2, красный = Wake). *Панель (d)*: длительность токена и вес классификатора.

The most notable finding is token T329 (ℓ=32 bins, 320 ms, w=+0.316): its ETA shows a sharp surface-negative deflection (≈−25 µV) followed by a positive dome (≈+10 µV) at ∼600 ms, which is morphologically consistent with a K-complex [31]. This token was discovered by BPE without any prior knowledge of sleep biomarker templates. The most discriminative Wake tokens are T62 and T63 (single-bin base tokens corresponding to the two largest adaptive amplitude bins, ≈+150–180 µV): their ETA shows narrow peak events consistent with EMG or movement transients.

Conclusions

BPE tokenisation for EEG classification has been systematically evaluated across six datasets, five paradigm types, six baselines, and several hyperparameter configurations. The central finding is a **paradigm-dependent effectiveness boundary**, which can be stated concisely:

> *BPE tokenisation is effective when a paradigm encodes discriminative information in the waveform (amplitude domain), and ineffective when the information lies in spectral power modulation (frequency domain).*

This boundary is empirically robust: it reproduces across all six datasets, five random seeds, and three classifier families. Its practical value is that knowledge of the paradigm's information coding mechanism determines BPE applicability *before* training any model.

Practical guidelines.

- **Amplitude paradigm** (sleep, P300). Use BPE_WindowedSeq_CNN or BPE_Hist_RF. Recommended settings are $B{\leq}64$, $V{\leq}512$, adaptive quantisation. Spatial BPE (k-means on amplitude snapshots) did not improve over channel-independent processing ($\kappa{=}0.693$ vs. 0.688 on Sleep-EDF).
- **Mixed paradigm** (mental arithmetic). Use PSD directly. The BPE+PSD ensemble (K7) adds <1 pp over PSD alone (68.3 vs. 68.2%), so BPE provides no complementary information once band-power is available.
- **Frequency paradigm** (motor imagery, SSVEP). Use PSD_LogReg or SSVEP_FFT; BPE will not close the gap. CSP+BPE yields a marginal gain ($+1$ pp) on small cohorts but does not scale; spectral baselines remain strongly preferred.
- **Interpretability**. BPE_Hist_LogReg provides sparse, human-readable token weights, making it the preferred choice when model transparency is required.

Limitations. BPE operates channel-independently; without an additional spatial step it does not capture inter-channel spatial patterns. The pipeline does not generalise across paradigms within a single model. The PhysioNet-MI exception (42.7%) requires a dedicated mechanistic investigation.

Future directions. Applying BPE to spectral representations (Fourier magnitude bins instead of amplitude bins) may close the gap for frequency-coded paradigms. Pretraining a foundation model on BPE tokens across multiple paradigms is another promising direction.

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